

1.1) Reálna časť:

Použijeme SL2 a SSS a definície skal. p.:

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} = \end{aligned}$$

$$= \|u\|^2 + \|v\|^2 + 2\operatorname{Re} \langle u, v \rangle, \text{ pretože } x + \bar{x} = 2\operatorname{Re}(x)$$

Podobne:

$$\begin{aligned} \|u-v\|^2 &= \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \overline{\langle u, v \rangle} = \|u\|^2 + \|v\|^2 - 2\operatorname{Re} \langle u, v \rangle \end{aligned}$$

$$\begin{aligned} \|u+v\|^2 - \|u-v\|^2 &= \|u\|^2 + \|v\|^2 + 2\operatorname{Re} \langle u, v \rangle - (\|u\|^2 + \|v\|^2 - 2\operatorname{Re} \langle u, v \rangle) \\ &= 4\operatorname{Re} \langle u, v \rangle \Rightarrow \operatorname{Re} \langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) \end{aligned}$$

Imaginárna časť:

$$\begin{aligned} \|u-iv\|^2 &= \langle u-iv, u-iv \rangle = \langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle \\ &= \|u\|^2 - i\langle u, v \rangle - i\overline{\langle u, v \rangle} + i\bar{i}\langle v, v \rangle \\ &= \|u\|^2 - i\langle u, v \rangle - i\overline{\langle u, v \rangle} + \|v\|^2 \\ &= \|u\|^2 + \|v\|^2 - 2i\operatorname{Im} \langle u, v \rangle \\ &= \|u\|^2 + \|v\|^2 + 2\operatorname{Im} \langle u, v \rangle \end{aligned}$$

analogicky: $\|u+iv\|^2 = \|u\|^2 + \|v\|^2 - 2\operatorname{Im} \langle u, v \rangle$

$$\|u-iv\|^2 - \|u+iv\|^2 = 4\operatorname{Im} \langle u, v \rangle$$

$$\Rightarrow \operatorname{Im} \langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$